



PORTFOLIO

Mr. Apiwat Nonut

IoT Engineer



List of Content



Education History



Project (Bachelor's Degree)



Thesis (Master's Degree)



Work experience



Certificate



Education History



Master's degree

Master of Engineering (Electrical Engineering)



Bachelor's degree

Bachelor of Engineering (Electrical Engineering)

GPA 3.35 (2nd Class Honors)



Project

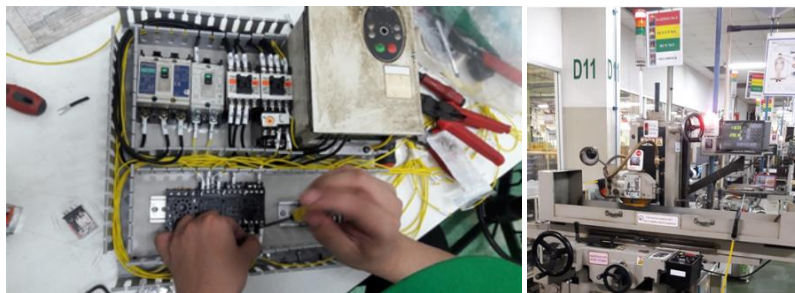
Summer internship at NHK SPRING Co.,Ltd.

Position: Maintenance Engineer



Project: Door opening system for using warehouse room by 4-digit password

Detail: We are conducting this project to practice using a Mitsubishi PLC and a pneumatic system to control a warehouse door opening system with a 4-digit password. The system allows up to four different passwords, which can be repeated or unique, each consisting of four digits. If the user does not enter a code within 10 seconds, the system will consider it an error, requiring the user to re-enter the code. After three consecutive incorrect attempts, an alert will be triggered. If the correct code is entered, the system will operate normally.



Project: Alarm system for grinding machines

Detail: We do this project to practice using Timer, Delay and circuit connections. The principle of operation is when the power button is pressed. There will be a red light on and the grinding machine will work. When the power off button is pressed, the yellow light will be on for 10 seconds to remind you to be careful and the green light will come on to show that the machine is safe.

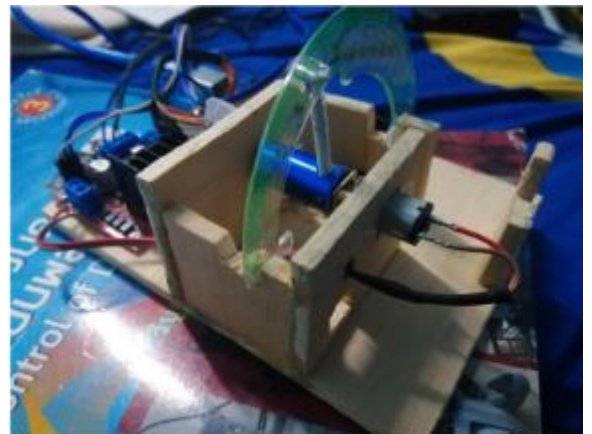
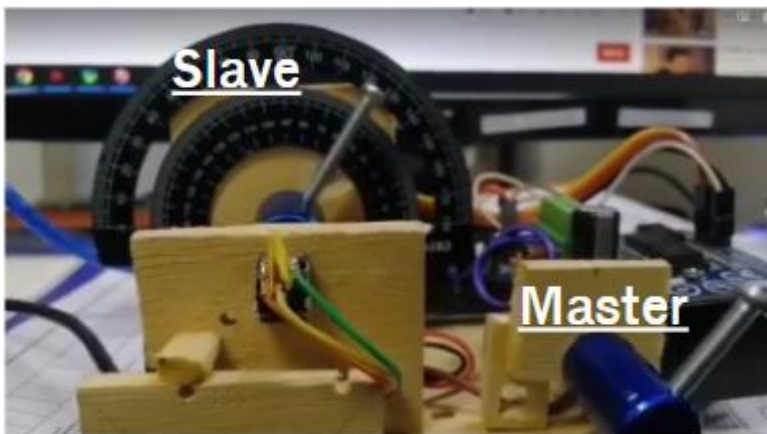
Bachelor's degree project

Electrical Engineering Khon Kaen University



Project: Servo Robot Arm

Detail: We made this project to practice using Solid works program to design 4 DOF robot arm with a 3D printer, and study the operation of the robot arm and apply Invest kinematics to control the operation and movement of the robot arm.



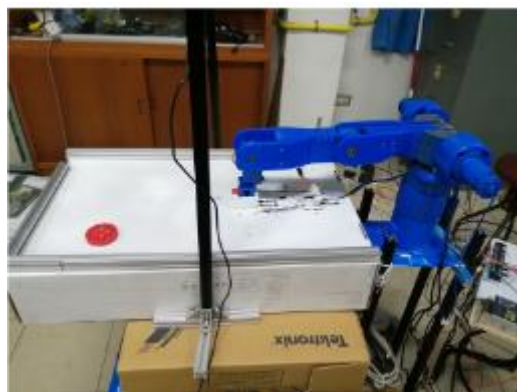
Project: Servo Robot Arm

Detail: We made this project to study the operation of a PID control system and test respond for each K_p K_i and K_d . It has two functions: Master or set point, which determines the desired position of the control system, and the second part, is the slave, which performs movements according to the command of the Master by using PID control system



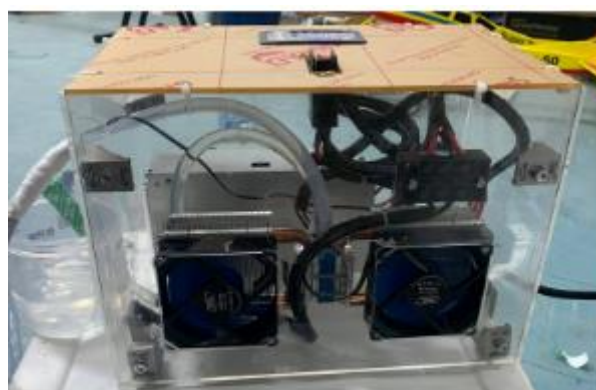
Project: PingPong Robot Arm

Detail: We made this project to study controlling a DC motor with a PID Control system and moving the end of factor to a specific position with LabVIEW programing. We study the image processing by LabVIEW for detect the position of ping pong ball and predict the final position of the ping pong ball so that the robot can counter the ping pong balls to competitor.



Project: Air Hockey Robot Arm

Detail: We apply the application in PingPong Robot for playing Air Hockey Robot with competitor.



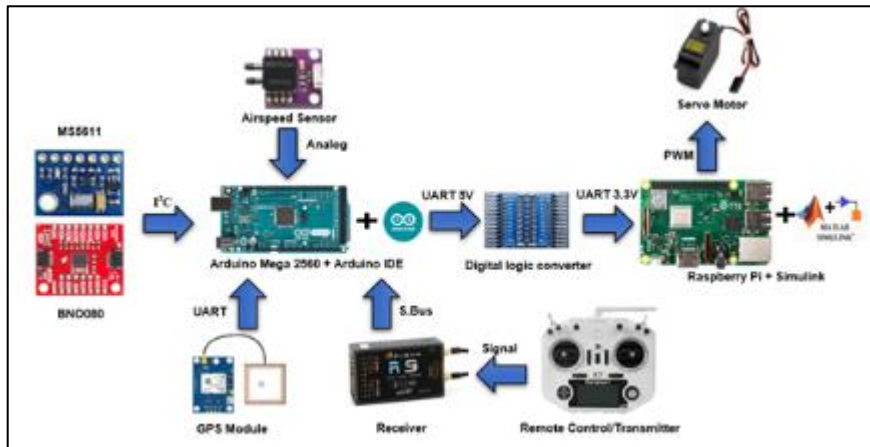
Project: Thermostat for brain injury patients with high body temperature.

(Cooperate with the Faculty of Nursing Khon Kaen University)



Thesis (Master's Degree)

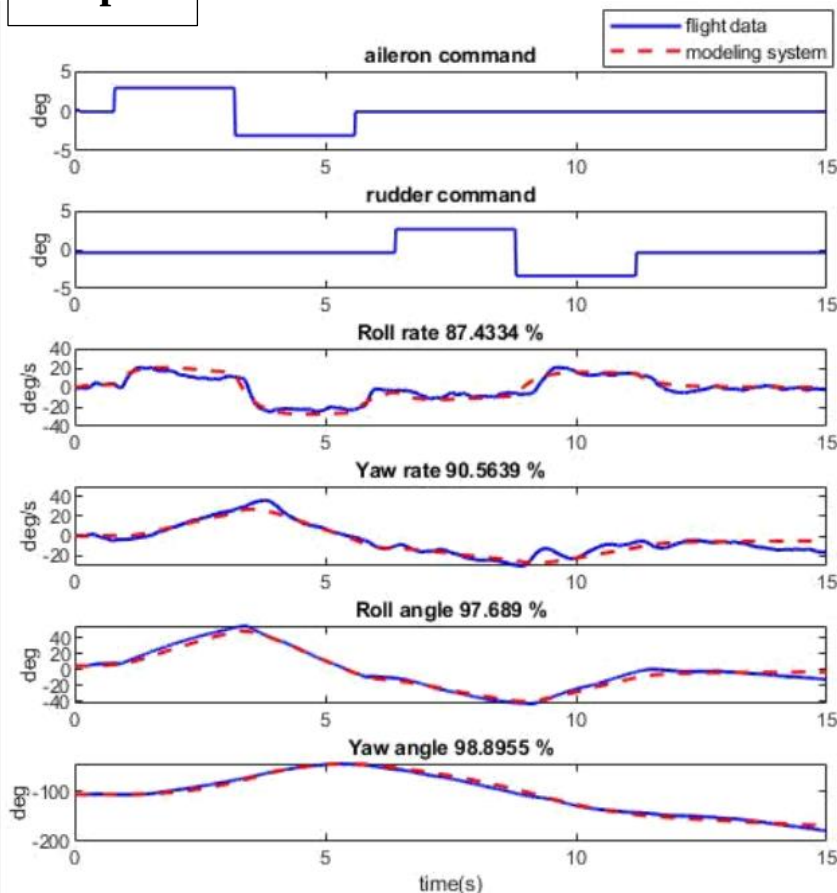
Topic: Mathematical Model and Control System Design for Fixed-Wing UAV (Drone) using Optimization Algorithm.



Control system design

Fixed-Wing Drone UAV

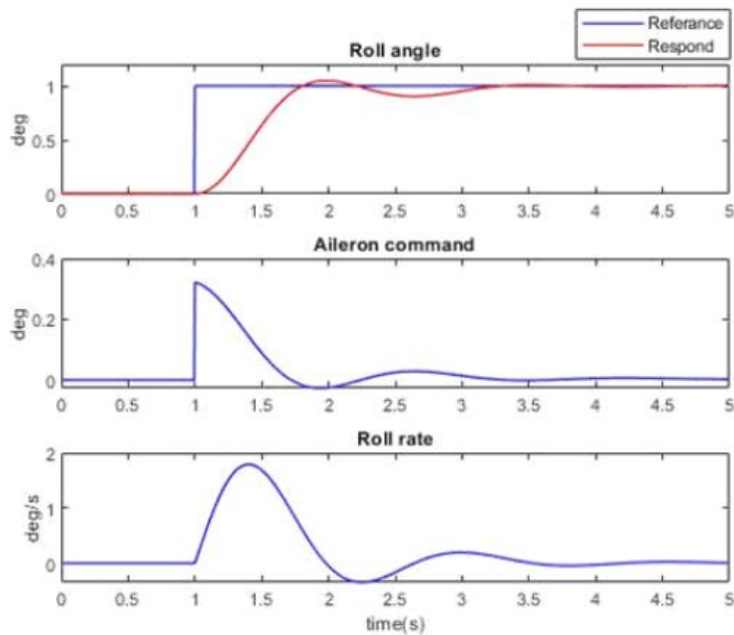
Step 1.



Find Mathematical Model

We find the Mathematical Model of Drone by excite system by input (2 graph at the top) and collecting real data (Blue Line) and find parameter in state space model which has the least error value between real data and mathematical model (red line) by using optimization technique. For the result, the accuracy more than 87% minimum and 98% for maximum

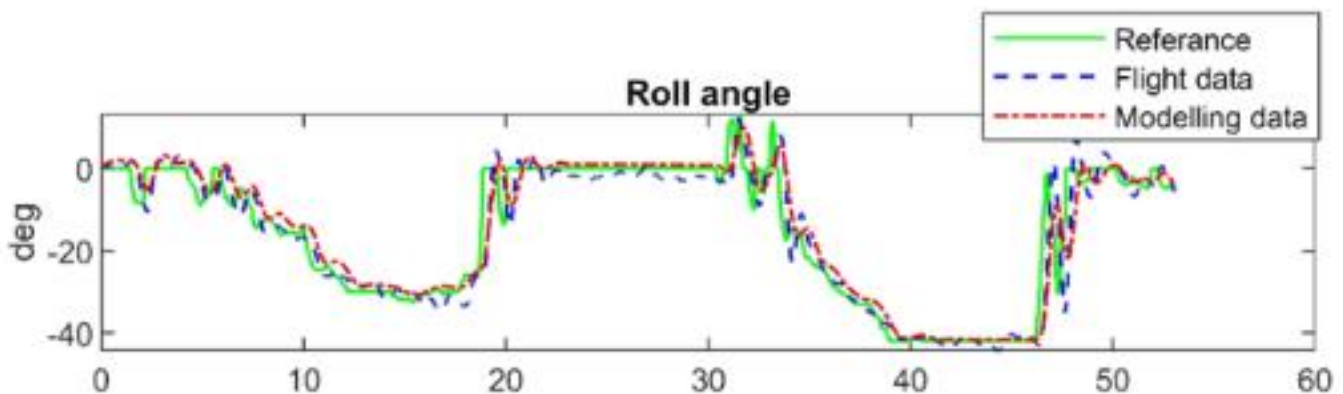
Step 2.



Design Control system

After we found the Mathematical Model of Drone from Step 1. Then, we designed PID control system by tuning the parameter K_p K_i and K_d by using Optimization technique

Step 3.



Test result from real flight compare with simulation flight

Assumption: A control system designed from a mathematical model can control the real drone with an error $< 10\%$.

Result: The Drone can track reference line (Green line) both flight data (blue line) and data from mathematical model (red line) with error $< 10\%$.

Thesis evaluation results >>> **Excellent**



Work experience

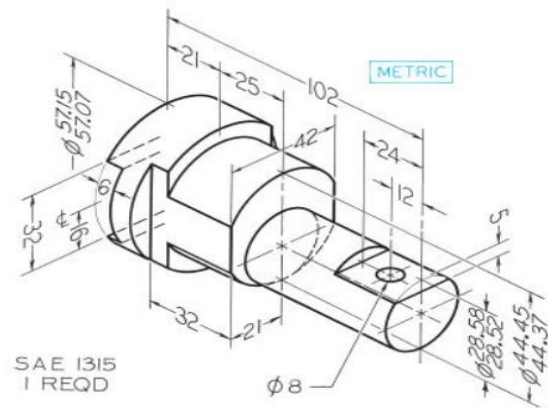
Year : 2019

Ck-Packaging, Phoenix Pulp and Paper Plc Ltd. & EN KKU

Position: Facilitator.

- Support the mechanic from SCG to learn engineering fundamentals and soft skills to solve problems within the factory.
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Year: 09/2021 – 08/2023

KUBOTA Research & Development Asia Co.,Ltd.

Position: IoT Engineer

- Study the feasibility of the market for Agricultural Machine.
 - Design new product concept for agriculture.
 - Product development planning.
 - Apply IoT Technology for development.
 - Develop Agricultural Machine with partners.
 - Performance test.
 - Others
-





Certificate



Certificate of Achievement

This is to certify that

Mr. Apiwat Nonut

has successfully completed

Mastering Indoor Vertical Farming for Cannabis (6 Hrs)

conducted on April 28, 2022.

A handwritten signature in blue ink.

Krisana Tamvimonol

CEO

Wangree Plant Factory Startup

In Cooperation with

WangreeFresh

A handwritten signature in blue ink.

Sirichai Kittivarapong, Ph.D.

Director

Career for the Future Academy

Certificate No. 20220824

P a v i n g T h e W a y f o r P r a c t i c a l S & T K n o w l e d g e

Certificate of Completion

This is to certify that

APIWAT NONUT

has successfully completed the course on

Test Case Design Techniques

on 8 August 2022



Mr. Prathan Dansakulcharoenkit
Instructor



Pattravadee Ploykitikoon, Ph.D.
Director, Software Park Thailand

No. 202200089



Certificate Verification

Software Park Thailand

National Science and Technology Development Agency
Ministry of Higher Education, Science, Research and Innovation



CERTIFICATE OF TRAINING COMPLETION

THIS CERTIFIES THAT

Apiwat Nonut

HAS BEEN SUCCESSFULLY COMPLETED THE TRAINING IN

BASIC STM32 MICROCONTROLLER PROGRAMMER

Pongpot Chalboonruang, Instructor

Sept 17, 2022

Date



IOT POW

วิทยาลัยอาชีวศึกษาปทุมธานี

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